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9 May 1980
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25 May 1966

MILITARY SPECIFICATION

TRUCK: BOOKMOBILE, COMMERCIAL, GASOLINE ENGINE DRIVEN, 17,000 POUNDS GVW, 4x2

This specification is approved for use by the US Army Tank-Automotive Materiel Readiness Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a bookmobile mounted on a forward control chassis. The vehicle shall be a gasoline engine driven, four wheel, two-wheel-driven truck having a gross vehicle weight (GVW) of not less than 17,000 pounds. Vehicles procured under this specification are commercial items which shall be warranted by the manufacturer as specified in procurement documents.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage: Vehicular, Ignition, Lighting and Starting.

MILITARY

MIL-E-24269 - Extinguisher, Fire, Carbon Dioxide, 15 Pound, Portable, Permanent Shutoff, Navy Shipboard Use.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Materiel Readiness Command, ATTN: DRSTA-GBS.1, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-2320

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STANDARDS
MILITARY
MIL-STD-1223 - Administrative Wheeled Vehicles Treatment, Painting, Rustproofing, Undercoating, Identification Marking, Data Plates and Warranty Notice Standards.

(Copies of specifications, standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies should reference the Code of Federal Regulations 40 CFR and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
NFPA No. 70-1978 - National Electrical Code

(Application for copies should be addressed to National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

STATE OF CALIFORNIA
Vehicle Code of California

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(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, California 95809.)

SOCIETY AUTOMOTIVE ENGINEERS, INC. (SAE)
SAE Standards and Recommended Practices

J245	- Engine Rating Code - Spark Ignition.
J537	- Storage Batteries.
J551	- Limits and Methods of Measurement of Radio Interference Characteristics of Vehicles and Devices (20-1000 MHz).
J552	- External Electromagnetic Radiation Suppressors.
J588	- Turn Signal Lamps.
J589	- Turn Signal Switch.
J688	- Truck Ability Prediction Procedure.

(SAE Standards and Recommended Practices are available from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.
Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 W. Market Street, Akron, OH 44313.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.6, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification. All chassis items shall be as represented in the chassis manufacturer's technical data, and special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be on file in the engineering offices of the procuring activity, prior to delivery of the items.

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3.1.1 Special requirements. In addition to standard vehicle and components specified in 3.1, vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Painting and marking. As specified by the procuring activity for the appropriate service (see 6.2), treatment, painting, identification marking, data plates and warranty notice shall be in accordance with MIL-STD-1223.

3.1.1.2 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with MIL-STD-1223.

3.1.1.3 Heavy duty cooling system. A heavy duty cooling system shall be furnished that shall maintain coolant at a temperature below the boiling point, with vehicle loaded to rated GVW and operated at 10,000 feet above sea level or in an ambient air temperature of not less than 125°F.

3.1.1.4 Coolant system. Coolant system shall include a de-aeration system, a surge tank or a coolant recovery reservoir of not less than two quart capacity.

3.1.1.5 Towing devices. Not less than two hooks, loops or pins, for towing the vehicle, shall be furnished on the front of the vehicle.

3.1.1.6 Rear end splash shields. Rear end splash shields shall extend from ends of steps to outside edge of body. All splash shield and mud flap installations, front and rear, shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. Vehicles shall comply with the Environmental Protection Agency Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

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3.2.3 Sound level. The interior sound level shall conform to Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to the Environmental Protection Agency Noise Emission Standards For Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. Curb weight shall include weight of chassis, with all attachments, accessories, and equipment; body; full complement of fuel, lubricants and coolant.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of curb weight, operator (weight computed at 175 pounds), and a payload to provide not less than the specified GVW.

3.2.6 Weight distribution. The distribution of gross vehicle weight shall be determined with the payload uniformly distributed over the load area.

3.2.7 Wheelbase. The wheelbase shall be not less than 150 inches.

3.2.8 Rating. Component and vehicle ratings shall not be raised to meet the requirements of this specification. Minimum GVW rating shall be not less than 17,000 pounds.

3.3 Performance.

3.3.1 Speeds and gradeability. High and low speed requirements shall be met with vehicle loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend 2.0 percent continuous grade at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive, and when multispeed axle is furnished, with the axle in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with engine operating at not less than 35 percent of recommended governed speed, and shall provide not more than 4.5 miles per hour.

3.3.1.3 Maximum geared speed. Maximum geared speed at engine governed speed shall be not less than 55 mph (minus 1 percent). Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

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$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction X tire factor}}$$

3.3.2 Service brakes. Service brakes shall control and hold vehicle, when loaded to its specified gross weight, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

3.4 Chassis components.

3.4.1 Engine. The vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine, with not less than six cylinders. The engine furnished shall provide the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of continuous, warranted operation on unleaded fuel. The rated net torque output of the engine shall be not less than 215 pound feet. Engine net torque ratings and net horsepower figures used in performance prediction calculations shall be determined in accordance with SAE J245.

3.4.1.1 Oil filter. A full flow type oil filter shall be furnished.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

3.4.2.1 Ignition system. An ignition system of 12-volt direct current (dc) potential shall be furnished. A generator or alternator of not less than 40 ampere rated capacity and which provides not less than 16 amperes dc output at normal engine idle speed shall be furnished.

3.4.2.2 Lighting. All vehicle lights, reflectors, and wiring shall be as specified herein and shall be in conformance with Federal Motor Carrier Safety Regulations 393.12, 393.19, 393.20, and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails or mounted on vehicle bumers.

3.4.2.3 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column.

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3.4.2.4 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity rating measured in accordance with SAE J537 shall not be less than 110 minutes. The cold cranking ampere rating at 0°F, measured in accordance with SAE J537, shall not be less than 450 amperes. Batteries shall be of the maintenance free type having the maintenance free characteristics listed in W-B-131.

3.4.2.5 Radio interference suppression. The vehicle shall be suppressed for radio interference in accordance with SAE J551. Suppressors used shall conform to SAE J552. Any body equipment emitting radio interference shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished. When equipped with an oil-bath type, oil capacity shall be not less than 1 quart.

3.4.3.2 Fuel tank. The fuel tank shall be the manufacturer's standard for the vehicle furnished with a capacity of not less than 30 gallons. When more than one tank is furnished, a selector valve connecting either tank to engine fuel intake shall be provided.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. The exhaust system shall discharge at the rear of the vehicle or to the side behind the rear wheels.

3.4.5 Automatic or semiautomatic transmission. An automatic or semiautomatic transmission shall be furnished. Automatic or semiautomatic transmission shall be of the fixed ratio type or infinitely variable type and shall provide continuous drive. The fixed ratio type shall include a hydraulic torque convertor and not less than three forward gear ratios. Normal driving range selector position shall provide not less than three gear ratios without movement of the selector. The infinitely variable type shall include a hydrostatic start up mode and a hydro-mechanical full speed mode with a control system to shift power flow from one mode to the other automatically. The driving range selector shall provide a normal drive position and a second, low speed, high power position.

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3.4.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. Chassis frame shall be manufacturer's standard for the vehicle furnished.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with vehicle loaded to specified GVW. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Vehicle shall be equipped with hydraulic, double-acting shock absorbers at the front wheels.

3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with vehicle loaded to specified GVW.

3.4.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with single front and dual rear wheels. Rims, and tire ratings shall conform to Tire and Rim Association recommendations, for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels on each vehicle.

3.4.10.1 Tires. Tires shall be tube or tubeless truck type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with vehicle loaded to specified GVW.

3.4.10.2 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

3.4.10.3 Carrier for spare tire assembly. Unless otherwise specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Carrier design shall enable removal or mounting of spare wheel assembly using only tools specified in 3.4.15.

3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished.

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3.4.10.5 Spare tire. When specified (see 6.2), an inflated spare tire mounted on the spare wheel or rim shall be furnished. The spare tire shall be of the same size, tread design and ply rating as the tires furnished on the vehicle.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.51.

3.4.11.1 Vacuum-hydraulic type. The vehicle shall be equipped with vacuum assisted, hydraulic actuated, four-wheel, service brakes.

3.4.12 Steering. Power steering shall be provided.

3.4.13 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by either air, vacuum with booster, or electric motor(s).

3.4.14 Bumpers. A standard front bumper shall be provided. A rear bumper shall be provided and shall conform to Federal Motor Carrier SAfety Regulation 393.86.

3.4.15 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging mounted tire assembly with the spare assembly (see 3.4.10.3), and shall include at least a jack, jack handle and wheel nut wrench. The jack shall be of such closed height as to permit its location under an axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of the wheel and tire assembly.

3.4.16 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to floor and to defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.17 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

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3.4.17.1 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. the following minimum equipment shall be furnished:

(a) Outside rearview mirrors, mounted on each side of body; each with at least 30 square inches of reflective area.
(b) Key-operated ignition switch.
(c) Ammeter, voltmeter or charging indicator.
(d) Fuel gage.
(e) Oil pressure gage or indicator.
(f) Engine temperature gage or indicator.
(g) Speedometer with recording odometer.
(h) Ash receptacle.
(i) Dual sunvisors.
(j) Driver's compartment ventilator other than window.

3.4.18 Horn. Manufacturer's standard electric horn shall be furnished.

3.5 Body. The body shall be an integral driver's compartment and enclosed body. The front section shall be the forward control type with cowl, windshield and side windows.

3.5.1 Body construction. The body shall be constructed of stretcher leveled sheet metal having a minimum gage thickness specified for carbon steel; high tensile steel utilized may be 2 gages lighter weight in accordance with U.S. Standard gages. All exterior panels shall be removable to facilitate body repair. Aluminum or magnesium alloys of equal strength may be furnished. Both sides of the body shall be protected with an all steel rub-rail located at the lower edge of the exterior panels. The rear of the body shall be protected with a full width bumper with wrap around ends (see 3.4.14).

3.5.2 Framing. Not less than six full-width crossmembers, including end sills, shall be furnished. Stub crossmembers, as required at the wheel housing to provide strength, shall be furnished. Body sides shall be reinforced for load support.

3.5.3 Service doors. The body shall incorporate a service entrance located forward on the curb side and a service exit located behind the rear wheels on the curb side. Inset step wells of safety tread steel or fiberglass reinforced plastic shall be furnished at each curb side entrance and exit. Hand rails shall be provided at the curb side service entrance and exit. The service doors shall be of the single, flush type not less than 26 inches wide by 84 inches high extending to the bottom of the step wells. Both of the curb side

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service doors shall be hung at the front with continuous type hinges covered with neoprene guards, fitted with manufacturer's standard door opening and locking arrangement, furnished with safety glass in the upper section, provided with push rails mounted on the inside, and equipped with an automatic door-closer and hold-back device. Doors shall be rubber sealed around all edges to provide a weather tight enclosure. Drip rails shall be installed above the doors.

3.5.4 Driver's doors. Manufacturer's standard automotive type cab door shall be installed on the road side. The door shall be fitted with ventilating type, safety glass in the upper section and shall be fitted with standard automotive hardware. The door shall be rubber sealed around all edges to provide a weather tight enclosure. A drip rail shall be provided above the door.

3.5.5 Rear window. A rear window shall be provided. The rear window shall be not less than 16 inches by 31 inches overall, fitted with single paned safety glass.

3.5.6 Storage compartment. An exterior storage compartment fitted with a weather tight door, embodying manufacturer's standard locking device, shall be located in the curb side or rear skirt. The exterior compartment shall be divided to house electrical equipment, and to store tools normally supplied with the vehicle.

3.5.7 Roof and exterior side panels. Roof and exterior side panels shall be stretcher leveled or formed construction with a minimum of 20 gage (0.0359 inch) sheet metal. All body seams and joints shall be dust and weather sealed in accordance with body building industry practice.

3.5.8 Skylights. Not less than two skylights shall be installed in the roof. Skylights shall be not less than 18 inches by 18 inches in size, and shall be fitted with translucent white butyrate dome. Skylights, consisting of a raised roof and transom type side windows in the raised portion, may be furnished.

3.5.9 Composite floor construction. Floor shall be multiple wood and metal construction to provide space for inserting insulating material. An all sheet metal under floor pan shall be an integral part of the flooring. The pan shall be secured to the bottom side of the floor cross sills to provide an insulation retainer and additional weather sealing. Wood floor joists, reinforced with steel channels, not less than 2 inches high, shall be installed on approximate 16 inch centers parallel to chassis cross members. The sub-floor shall be all

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wood construction secured to the floor joists and shall extend the full length and width of load space. The sub-floor shall be not less than 3/4 inch thick planking, 3/4 inch laminated plywood, or shiplap, and shall be covered with 1/4 inch plywood with a minimum of joints. A vapor barrier of tar paper or metallic foil shall be inserted between the sub-floor and plywood. The floor shall be insulated with not less than 2 inches of fiberglass batts.

3.5.10 Exterior wheelhousings. The exterior wheelhousings shall be constructed of not less than 16 gage steel and shall be properly sealed to prevent dust and water seepage. The exterior wheelhousings shall be designed to provide clearance for the dual rear tires under all load conditions.

3.5.11 Interior body dimensions. Inside body dimensions shall be not less than 198 inches long from back of driver's seat to rear interior paneling; 87 inches wide inside of interior paneling exclusive of wheelhousings; and 84 inches high from floor to ceiling, at center of body.

3.5.12 Interior body construction. Interior body sides and rear shall be paneled with 1/4 inch finished plywood of color and decoration to match shelving. Exposed sides shall be free from defects and sanded smooth. The ceiling shall be lined with sheet metal not less than 20 gage (0.0359 inch) thick. All interior panels shall be removable to facilitate body repair. Body sides, rear and roof shall be insulated with not less than 2 inches of fiberglass batts. Inside wheelhousings shall be of square wood construction, sealed against dust and moisture, and fully insulated.

3.5.13 Driver's cab interior. Cab headliner shall be sheet metal, 1/8 inch tempered masonite, 1/4 inch finished plywood with the exposed side free from defects and sanded smooth, or a one-piece fiberglass liner providing a smooth finished surface. Cab roof, floor, and cowl shall be insulated with not less than 1 inch thick fiberglass batts. Cab side walls shall be carpeted. The cab shall be furnished with a fully adjustable, bucket type driver's seat mounted on a swivel base. Seat shall be capable of being locked in standard driving position, or revolved 360 degrees for use at the front service and loan desk. Manufacturer's standard fastening for one pair of seat belts shall be installed on the vehicle.

3.5.14 Floor covering and cove base. The floor shall be covered with high quality battleship linoleum or vinyl tile. Linoleum shall be installed over not less than 14 pound felt base. Molding shall be

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provided at the base of shelving, desks, lockers, wheelboxes, and other projections. Pattern and colors furnished shall be as specified by the procuring activity (see 6.2).

3.5.15 Interior compartments. A clothes closet shall be installed on the left side of the vehicle, behind and adjacent to the front service and loan desks. Outside dimensions of the closet shall be not les than 10 1/2 inches deep, 24 inches wide, and 77 inches high. A locker shall be furnished with a full length door fitted with lock and slam type catch, equipped with a shelf at the top, and provided with a hanger bar. Adjoining the clothes locker, an auxiliary compartment, with space for a vacuum-insulated jug of 5 gallons capacity shall be installed. The compartment shall provide for storage of drinking cups and miscellaneous items.

3.5.16 Shelving. Shelving shall be manufacturer's current standard, adjustable bookmobile type and shall be computed at the rate of 8 to 10 books per foot of shelf length. A bank of shelves shall be furnished on each, left and right side, of the vehicle; and provided on each end of reference desk at the rear of the vehicle. The banks of shelves on each side of the vehicle shall be sloped 15 degrees; and the banks of shelves at the rear of the vehicle shall be sloped 20 degrees. The shelves shall be not less than 7 inches deep and 10 1/2 inches clear height; shall be fitted with 90 degree backing strips to enable books to stand squarely and resist unnecessary wear; and shall be supported by a pair of surface mounted pilaster standards at each end. The shelves and standards shall be built of kiln dried 3/4 inch finished lumber; or constructed of 3/4 inch finished plywood with the exposed surfaces free of defects and sanded smooth; or fabricated of sheet metal. No rubber matting shall be applied on the shelves.

3.5.17 Magazine racks. Two magazine racks shall be furnished. One rack shall be provided on each end of the bank of bookshelves adjacent to the entrance and exit service doors.

3.5.18 Bulletin board. One bulletin board, constructed of 1/4 inch sheet cork mounted on a plywood base and framed, shall be furnished; and shall be mounted above the rear window and reference desk.

3.5.19 Desks. The vehicle shall be furnished with a loan desk located behind the driver's seat and shall be provided with a reference desk located below the rear window and between bookshelves in either corner. Both of the desks shall be constructed of sheet metal or not less than 3/4 inch finished lumber or 3/4 inch finished plywood with exposed surfaces free from defects and sanded smooth. The top of each

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desk shall be covered with Formica of a color and pattern to match the decor of the interior of the vehicle. The top of the loan desk shall be not less than 18 inches deep by 48 inches wide and shall be approximately 35 inches above the floor. The loan desk shall be furnished with a drawer not less than 3 inches deep and shall be provided with a storage compartment for miscellaneous library supplies. The top of the reference desk shall be not less than 15 inches deep by 48 inches wide and shall be approximately 39 inches above the floor. The underneath section of the reference desk shall be a storage cabinet for records, films and other miscellaneous items. The reference desk storage cabinet shall be provided with adjustable shelves and shall be equipped with double doors fitted with locks, handles and catches.

3.5.20 Interior lighting and service outlets. The interior of the body shall be equipped with 2 continuous rows of 120-volt, instant starting type fluorescent fixtures complete with starters, 25 watt 24 inch tubes, and plastic shades. The fluorescent fixtures shall be mounted flush against the ceiling and approximately 15 inches from each bank of side shelving. Control switches for the lights shall be mounted near the loan service desk. A plug-in outlet shall be mounted for each air conditioner (see 3.5.22) on either side of the vehicle. A duplex convenience receptacle shall be provided near the front loan service desk. A similar convenience outlet shall be furnished above the rear reference desk.

3.5.21 Heating. The vehicle shall be furnished with three, manufacturer's standard commercial resistance type, 240 volt, electric heaters. Each heater shall be equipped with a circulating fan and an automatic temperature control. The rating per unit of two heaters shall be 13,660 British thermal units (Btu) per hour at 4000 watts, and the capacity of the third unit shall be 5,122 Btu per hour at 1500 watts. One 4000 watt heater shall be mounted underneath the bookshelves and adjacent to the front service entrance. The second 4000 watt heater shall be mounted underneath the bookshelves and adjacent to the rear service exit. The 1500 watt heater shall be mounted underneath the front loan service desk.

3.5.22 Air conditioning. The vehicle shall be provided with two, manufactuer's standard commercial type, air conditioners. The rating of each unit shall be 1 hp at 120 volts, and the capacity per unit shall be not less than 9,700 Btu per hour. One air conditioning unit shall be installed at ceiling height above and toward the front of the bank of bookshelves on the left side of the vehicle. The second unit shall be installed at ceiling height above and toward the rear of the bank of bookshelves on the left side of the vehicle. Both air conditioners shall be equipped with thermostats for automatic temperature control.

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3.5.23 Electrical power systems. The interior of the body shall be wired in accordance with the requirements of the National Electrical Code, NFPA No. 70-1978, for mobile homes, and electrical wiring code of the state in which the vehicle is to operate. Five, 120-volt power circuits shall be installed. Each row of lights shall be on a separate circuit, each air conditioner shall be on an individual circuit, and both of the convenience outlets shal be on the same circuit. Three 240-volt power circuits shall be installed. Each heater shall be provided with a separate circuit. Each individual circuit shall be protected with approved type fuses or multi-breaker panel of applicable rating. The fuse disconnect switch box or multi-breaker panel for both panel systems, 120-volt and 240-volt, shall be mounted in the cab above the driver's door. The wiring for both systems shall be installed in flexible or rigid conduit as applicable conditions may warrant. The vehicle shall be furnished with two weather proof shoreline receptacles for power. One outside entry receptacle shall be mounted on the left side of body; and the other entry receptacle shall be installed in the reel compartment. Provision shall be made to disconnect either receptacle when not in use. Not less than 100 feet of No. 8, three conductor, neoprene covered extension cable shall be furnished on a reel in the skirt compartment.

3.5.24 Public address system. The public address system shall consist of one combination amplifier and turntable, one microphone, two loudspeakers and unit inter-connecting cords, plugs, and other necessary items. Specific specifications of components shall be as follows:

AMPLIFIER AND TURNTABLE:

Power output: 20 watts.
Frequency response: Plus or minus 2 decibels (db) at 20 to 18,000 hertz (Hz).
Input: Provision for phonograph and microphone.
Turntable: Three speeds, 33 1/3, 45 and 78 revolutions per minute (rpm).
Amplifier shall be rack mounted in a suitable location to permit ready accessibility and provided with shock mounts to withstand road shock and vibrations.
Bogen #CHA62DY, or equivalent.

MICROPHONE:

Type: Dynamic, unidirectional, furnished with desk stand, complete with cable and plug assemblies, Shure microphone 55S and Shure Model S36A stand, or equivalent.

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LOUDSPEAKER:

Power handling capacity: 20 watts.
Frequency response: Plus or minus 5 db at 175 to 12,000 Hz.
Type: Outdoor re-entrant Electro-Voice Model 848 with 878 matching transformer, or equivalent.

Speakers shall be mounted horizontally on the forward and rear sections of the bookmobile so as to project maximum sound in the forward and rear direction of the bookmobile while in motion. A suitable metallic guard shall be provided for the purpose of protecting the speakers from damage due to engagement with overhanging branches or other obstacles encountered while in travel.

3.5.25 Clock. The vehicle shall be furnished with an 8 day, spring wound, wall mounting type clock. The clock shall be located near the rear reference desk.

3.5.26 Fire extinguisher. The vehicle shall be provided with a 15 pound carbon-dioxide, fire extinguisher conforming to MIL-E-24269. Mounting for the fire extinguisher shall be in a readily accessible area.

3.6 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, contractor shall service and adjust each vehicle for operational use including at least the following: focusing of lights; adjustment of the engine, electrical and brake systems; filling and charging of battery; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; and servicing of cooling system with a solution of ethylene glycol and water in equal parts by volume.

3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted, and riveted construction utilized shall be in accordance with the highest standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements

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specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformity with the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Vehicle weight. The first production vehicle shall be weighed to determine curb weight and distribution of curb weight on front and rear axles. The imposed loading on front and rear axles will be computed using the curb weight, the operator weight at 175 pounds, and the payload (see 3.2.4) to provide the specified GVW. Calculated imposed loads on front and rear axles will be utilized to ascertain that the suspension, axles and tires furnished are of adequate capacity to meet contract requirements.

4.3.2 Road test. The first production vehicle shall be examined and road tested (without payload) by the contractor to assure that the vehicle will operate in accordance with the contract requirements.

4.3.3 Radio frequency suppression verification. The vehicle manufacturer shall furnish, with the questionnaire (commercial vehicle engineering data form), a description of all interference suppression devices used to suppress vehicle electromagnetic radiation in accordance with SAE J551. Data shall be sufficient for visual determination that all required suppression devices are installed on the vehicle.

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4.3.4 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.3.5 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.6 has been accomplished.

5. PACKAGING

5.1 Vehicle processing. Vehicle shall be processed for shipment, from manufacturer's plant to initial receiving activity, in accordance with manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. Bookmobiles covered by this specification are intended to provide library service to off-post sites; to be operated by a male or female librarian over improved or unimproved roads and highways, under all conditions of weather and climate encountered within the continental limits of the United States and at various altitudes and grades which may be encountered on the main roads, highways and access roads within the continental limits of the United States.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Identification of the appropriate service for painting and marking (see 3.1.1.1).
(c) Rustproofing, if required (see 3.1.1.2).
(d) If carrier for spare tire is not required (see 3.4.10.3).
(e) A spare wheel or rim, if required (see 3.4.10.4).

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(f) A spare tire, if required (see 3.4.10.5).
(g) Tools, if required (see 3.4.15).
(h) Floor covering pattern and color (see 3.5.14).

6.3 Performance prediction. Completed copies of SAE Truck Ability Prediction form and computation for low speed requirement for each vehicle model furnished under contract shall be submitted as specified in the contract. The SAE Work Sheet Item 1 shall include vehicle model number, wheelbase, engine model number, and vehicle type. Unless other conditions are cited in the contract, computations shall be made for normal atmospheric pressure, normal ambient air temperature, and still, dry, air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1 - Tire Factor. This factor must relate to the size of tire furnished by the contractor in accordance with this specification.
Table 2 - Altitude Factor. 1.00
Table 3 - Rolling Factor. 1.613
Table 4 - Area Factor. 0.173
Table 5 - Velocity Factor. 250.0
Table 6 - Altitude Factor. 1.00
Table 7 - Chassis Friction Horsepower. Use vehicle GVW (to nearest, higher, 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 50 mph geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8 - Grade Factor. 0.75
Table 8A - Correction Factor. Not required.
Table 9 - Road Factor. 0.0.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-T-45305B(AT) TRUCK: BOOKMOBILE, COMMERCIAL, GASOLINE ENGINE DRIVEN, 17,000 POUNDS GVW, 4x2

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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